

# Minime CI Workflow Architecture (Vertical Pipeline)

Unified GitHub Actions Build System — .github/workflows/build.yml [\[Shared\]](#)

Legend: [\[Shared\]](#) = Dual-distro / shared across targets | [\[Specific\]](#) = Single-distro, board, or libc specific

## 1. Trigger / Entrypoint

- **Workflow File:** .github/workflows/build.yml [\[Shared\]](#)
- **Triggers:** Push to main, Pull Request, or workflow\_dispatch

↓ (Triggers parallel Tier 1 jobs)

## 2. Parallel Tier 1 Builds (Bootloader & UIs)

| 2A. Bootloader Build                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2B. UI Build (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2C. UI Build (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>build-bootloader</b> (AMD64) <ul style="list-style-type: none"><li>• <b>Workflow Job:</b> build-bootloader <a href="#">[Shared]</a></li><li>• <b>Runner:</b> ubuntu-latest</li><li>• <b>Container:</b> minime-glibc:latest <a href="#">[Shared]</a></li><li>• <b>Script:</b> scripts/build-bootloader.sh <a href="#">[Shared]</a></li><li>• <b>Cache Key:</b> bootloader-* (hash of uboot/**)</li><li>• <b>Configs &amp; Patches:</b><ul style="list-style-type: none"><li>- minime/uboot/config/uboot.config <a href="#">[Shared]</a></li><li>- minime/uboot/config/ddr3.defconfig <a href="#">[Specific]</a> (H700)</li><li>- minime/uboot/out/rk3566/rkbin/ <a href="#">[Specific]</a> (RK3566)</li></ul></li><li>• <b>GH Artifact:</b> bootloader-out</li></ul> | <b>build-ui (musl)</b> (ARM64) <ul style="list-style-type: none"><li>• <b>Workflow Job:</b> build-ui (musl) <a href="#">[Shared]</a></li><li>• <b>Runner:</b> ubuntu-24.04-arm</li><li>• <b>Container:</b> minime-musl:latest <a href="#">[Shared]</a></li><li>• <b>Script:</b> scripts/build-ui.sh minui allium musl <a href="#">[Shared]</a></li><li>• <b>Cache Key:</b> ui-musl-* (hash of ui/**)</li><li>• <b>Submodules:</b><ul style="list-style-type: none"><li>- minime/ui/minui/ <a href="#">[Shared]</a> (dual musl/glibc)</li><li>- minime/ui/allium/ <a href="#">[Shared]</a> (dual musl/glibc)</li></ul></li><li>• <b>GH Artifact:</b> ui-musl<ul style="list-style-type: none"><li>■ minui-musl-aarch64.tar.xz</li><li>■ allium-musl-aarch64.tar.xz</li></ul></li></ul> | <b>build-ui (glibc)</b> (AMD64) <ul style="list-style-type: none"><li>• <b>Workflow Job:</b> build-ui (glibc) <a href="#">[Shared]</a></li><li>• <b>Runner:</b> ubuntu-latest</li><li>• <b>Container:</b> minime-glibc:latest <a href="#">[Shared]</a></li><li>• <b>Script:</b> scripts/build-ui.sh minui allium glibc <a href="#">[Shared]</a></li><li>• <b>Cache Key:</b> ui-glibc-* (hash of ui/**)</li><li>• <b>Submodules:</b><ul style="list-style-type: none"><li>- minime/ui/minui/ <a href="#">[Shared]</a> (dual musl/glibc)</li><li>- minime/ui/allium/ <a href="#">[Shared]</a> (dual musl/glibc)</li></ul></li><li>• <b>GH Artifact:</b> ui-glibc<ul style="list-style-type: none"><li>■ minui-glibc-aarch64.tar.xz</li><li>■ allium-glibc-aarch64.tar.xz</li></ul></li></ul> |

↓ (Uploads artifacts to GH & Passes to Tier 2)

## 3. Artifact Transfer & Caching Layer

- **Actions Used:** actions/download-artifact@v4 [\[Shared\]](#), actions/cache/restore@v4 [\[Shared\]](#)
- **Artifacts Downloaded:** bootloader-out + matching UI artifact (ui-musl for Alpine, ui-glibc for Buildroot)
- **Tier 1 Output Caches:** Bootloader Cache (bootloader-\*), UI Cache (ui-musl-\*, ui-glibc-\*)
- **Tier 2 OS Caches:** CCache (~/.alpine-ccache, ~/.buildroot-ccache) + DL Cache (~/.alpine-dl, ~/.buildroot-dl) [\[Specific\]](#)

↓ (Fans out to 6 parallel matrix jobs)

## 4. Target Component Compilation (build.sh)

| 4A. Alpine Component Builder (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4B. Buildroot Component Builder (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Alpine (musl) Component Builder (3 Jobs)</b> <ul style="list-style-type: none"><li>• <b>Command:</b> make -C minime/targets/alpine components BOARD=&lt;board&gt; <a href="#">[Specific]</a></li><li>• <b>Builder Script:</b> minime/targets/alpine/scripts/build.sh <a href="#">[Specific]</a></li><li>• <b>Inputs:</b> minime/boards/&lt;board&gt;/ <a href="#">[Specific]</a>, minime/boards/common/ <a href="#">[Shared]</a></li><li>• <b>Produced Artifacts:</b><ul style="list-style-type: none"><li>■ system.erofs (EROFS rootfs: kernel, modules, GPU drivers)</li></ul></li></ul> | <b>Buildroot (glibc) Component Builder (3 Jobs)</b> <ul style="list-style-type: none"><li>• <b>Command:</b> make -C minime/targets/buildroot components BOARD=&lt;board&gt; <a href="#">[Specific]</a></li><li>• <b>Builder Script:</b> minime/targets/buildroot/scripts/build.sh <a href="#">[Specific]</a></li><li>• <b>Inputs:</b> minime/boards/&lt;board&gt;/ <a href="#">[Specific]</a>, minime/boards/common/ <a href="#">[Shared]</a></li><li>• <b>Produced Artifacts:</b><ul style="list-style-type: none"><li>■ system.erofs (EROFS rootfs: kernel, modules, GPU drivers)</li></ul></li></ul> |

↓ (Passes system.erofs to Packaging)

## 5. Image Assembly & Packaging (mkimage.sh / mkupdate.sh)

| 5A. Alpine Image Assembly (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5B. Buildroot Image Assembly (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Alpine Image &amp; Update Packager (musl)</b> <ul style="list-style-type: none"><li>• <b>Commands:</b><ul style="list-style-type: none"><li>- make image BOARD=&lt;board&gt; UI=minui allium <a href="#">[Specific]</a></li><li>- make update BOARD=&lt;board&gt; UI=minui allium <a href="#">[Specific]</a></li></ul></li><li>• <b>Packaging Scripts:</b><ul style="list-style-type: none"><li>- minime/build/genassets.sh <a href="#">[Shared]</a></li><li>- minime/build/mkimage.sh <a href="#">[Shared]</a></li><li>- minime/build/mkupdate.sh <a href="#">[Shared]</a></li></ul></li><li>• <b>Produced Release Archives:</b><ul style="list-style-type: none"><li>■ *.img.xz (Raw SD card flashable image)</li><li>■ *.tar.xz (OTA update package)</li></ul></li></ul> | <b>Buildroot Image &amp; Update Packager (glibc)</b> <ul style="list-style-type: none"><li>• <b>Commands:</b><ul style="list-style-type: none"><li>- make image BOARD=&lt;board&gt; UI=minui allium <a href="#">[Specific]</a></li><li>- make update BOARD=&lt;board&gt; UI=minui allium <a href="#">[Specific]</a></li></ul></li><li>• <b>Packaging Scripts:</b><ul style="list-style-type: none"><li>- minime/build/genassets.sh <a href="#">[Shared]</a></li><li>- minime/build/mkimage.sh <a href="#">[Shared]</a></li><li>- minime/build/mkupdate.sh <a href="#">[Shared]</a></li></ul></li><li>• <b>Produced Release Archives:</b><ul style="list-style-type: none"><li>■ *.img.xz (Raw SD card flashable image)</li><li>■ *.tar.xz (OTA update package)</li></ul></li></ul> |

↓ (Uploads output archives to GitHub Releases)

## 6. Release & Delivery

- **Workflow Step:** Upload to Testing Release in .github/workflows/build.yml [\[Shared\]](#)
- **Tool / CLI:** gh release upload testing /\*.img.xz /\*.tar.xz --clobber [\[Shared\]](#)
- **Destination:** GitHub Releases — **testing** tag